

Getting Started with DORA

Connecting to your vehicle, and a walkthrough of the core tuning workflows.

This guide covers hardware setup, connecting DORA to a vehicle over OBD-II, and step-by-step instructions for live monitoring, data logging, diagnostics, dyno testing, map editing, AI analysis, and ECU flashing.

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1. What You'll Need

DORA runs as a desktop application (Windows) and can operate in two modes: **Simulator** (no hardware — built-in synthetic engine data, useful for practice and offline map work) and **Live OBD-II** (connected to a real vehicle).

For live monitoring, logging & diagnostics

- An ELM327-compatible OBD-II adapter (USB or Bluetooth).
- A USB cable (if using a wired adapter) or a paired Bluetooth connection.
- The vehicle's OBD-II port — usually under the dash, driver's side, within reach of the seat.

Additionally, for ECU flashing

- A supported K-line (KWP2000) or CAN (UDS) interface / J2534 pass-thru device.
- The tuned ROM (.bin) file you intend to write.
- A battery charger/maintainer connected to the vehicle — flashing must never run on a battery that can sag mid-write.

2. Connecting DORA to Your Vehicle

- 1 Plug the OBD-II adapter into the vehicle's OBD-II port, and connect it to your PC (USB) or pair it over Bluetooth.
- 2 Turn the vehicle's ignition to **ON** (the engine does not need to be running for the adapter to establish a connection).
- 3 Launch DORA. From a source checkout: `cd ecu_tune/ecu_tuner && python main.py` — or run the installed application.
- 4 In the app header, click the **connection pill** (shows "Simulator" by default) to open the **Connect to Vehicle** dialog.
- 5 Under **Live OBD-II**, click **Refresh** to list available COM ports, then select the port matching your adapter.
- 6 Set the **Baud** rate — 38400 is the default for most ELM327 adapters; try 9600, 57600, or 115200 if the connection fails.
- 7 Click **Connect**. DORA will attempt the handshake in the background — this can take a few seconds.
- 8 On success the dialog shows "Connected on COMx" and the header pill turns green: "Live OBD-II — COMx". All panels (Live Monitor, Data Logging, Fault Codes, etc.) now read from the real vehicle.
- 9 If the connection fails, you'll see an inline error. Double-check the adapter is seated, the port is correct, and the ignition is on — or click **Use Simulator** to keep working offline.
- 10 To end a live session, reopen the connection dialog and click **Disconnect** — DORA falls back to the simulator automatically so panels never go blank.

3. Live Monitor — Real-Time Gauges

- 1 In the sidebar, under **MONITORING**, click **Live Monitor**.
- 2 Gauges populate from whichever source is active — Simulator or Live OBD-II — showing RPM, coolant temp, intake temp, MAF, throttle position, boost, short/long-term fuel trim, O2 voltage, and knock retard.
- 3 Press **C** to open the **Gauge Editor** and rearrange, resize, or swap gauges in the grid by drag-and-drop.
- 4 Press **K** from the dashboard to start trigger-based logging (RPM or a channel threshold) — see Section 4.

4. Data Logging

- 1 With a connection active (Live or Simulator), open **Live Monitor** and press **K** to arm trigger logging, or open **Log Viewer** from the sidebar (MONITORING) to manage recorded sessions.
- 2 Set a trigger — e.g. RPM above a threshold, or a specific channel crossing a value — so logging starts automatically once conditions are met (useful for WOT pulls).
- 3 Logged sessions are written as CSV files, which you can reorder, filter, and review channel-by-channel inside Log Viewer (drag-and-drop column reordering supported).
- 4 Use **Fill From Log** (ANALYSIS) to populate map cells directly from a recorded log, or **Log Compare** to overlay two sessions.

5. Reading & Clearing Fault Codes

- 1 In the sidebar, under **WORKFLOW**, open **Fault Codes** (or **Fault Codes+** for the expanded view).
- 2 Click **Scan** to pull current DTCs from the ECU over the active connection.
- 3 Each code resolves against DORA's built-in 56-code DTC library for a plain-language description.
- 4 Review, then clear codes from the same panel once the underlying issue is addressed.

6. Dyno Test

- 1 In the sidebar, under **MONITORING**, open **Dyno Test**.
- 2 Run a pull — live (vehicle on rollers, connected) or simulated, using the built-in engine model.
- 3 Results are corrected to SAE or DIN standard automatically.
- 4 Export the pull as a PDF report, or overlay it against a previous run in **Dyno Overlay** (ANALYSIS).

7. Map Editor

- 1 In the sidebar, under **TUNING**, open **Map Editor**.
- 2 Open a ROM (via ROM Identify / ROM Library) to load its maps.
- 3 Edit cells directly, or use the 3D surface view, smoothing, and region fill/extrapolate/blend tools for broader changes.
- 4 Compare against another map or a known-good tune from the library before committing.
- 5 Every change is tracked with full undo/redo, and every write is checked by the **Safety Guard** before it can reach the ECU (see Section 11).

8. AI Analysis

- 1 In the sidebar, under **AI**, open **AI Analysis** (or **AI Analysis+** for expanded output).
- 2 Run analysis against a loaded map or log — DORA flags knock risk, fuel trim drift, and tuning opportunities directly on the map.
- 3 Use the natural-language explainer to ask what a specific cell or region change actually does before applying it.

9. Flashing the ECU

In the sidebar, under **ECU**, open **ECU Flash** to launch the step-by-step Flash Wizard.

- 1 Protocol** — choose KWP2000 (serial K-line) or UDS (CAN), matching your vehicle and interface.
- 2 Connection** — enter the port/channel and, for CAN, the TX/RX arbitration IDs and bitrate (or baud rate for K-line).
- 3 ROM File** — browse to the tuned binary you intend to write.
- 4 Checksum** — the wizard verifies the ROM's checksum automatically; flashing is blocked if it doesn't validate.
- 5 Flash** — the ROM streams to the ECU with a live sector-by-sector progress bar. Do not disconnect the adapter, turn off the ignition, or lose power to the vehicle during this step.
- 6 Done** — a success or error summary is shown, and the flash is recorded to the flash history log for later rollback reference.

Before you flash

- Confirm a battery charger/maintainer is connected — a voltage sag mid-write can brick the ECU.
- Keep the original ROM backed up (ROM History keeps automatic snapshots you can roll back to).
- Never flash with the engine running.

10. Troubleshooting

Symptom	Likely cause / fix
No ports listed in the connection dialog	Adapter driver isn't installed, or the adapter isn't plugged in / paired. Reconnect it and click Refresh.
"Could not connect" on the selected port	Check the ignition is ON, the correct port is selected, and try a different baud rate.
Gauges show no movement	Confirm the header pill reads "Live OBD-II", not "Simulator" — if it still reads Simulator, the connect step didn't complete.
Unsure if it's a software or hardware issue	Switch to Simulator mode first to confirm the app and panels work, then isolate the hardware side.

11. Safety Notes

Built-in protection

- The Safety Guard checks every map change against per-cell and total-cells-changed limits before it can be written — this check cannot be bypassed.
- Live sensor data (boost, coolant temp, EGT, knock) is continuously validated against hard limits while monitoring.

Your responsibility

- Always keep the vehicle's battery on a maintainer during any ECU write.
- Never disconnect the OBD-II adapter or interrupt power mid-flash.
- Keep a verified-good backup of the original ROM before tuning a vehicle for the first time.